Phillip W. Yoon

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Education

Hunter College, City University of New York 2025–2027 (expected)
M.A. in Mathematics. *New York, NY*

University of Pennsylvania, Graduate School of Education 2022–2024
M.S.Ed. in Independent School Teaching Residency *Philadelphia, PA*

Northern Illinois University 2021–2022
Graduate (M.A.) coursework in philosophy, 18 credits (non-degree) *DeKalb, IL*

Johns Hopkins University 2017–2021
B.A. in Mathematics and Philosophy, General Honors. *Baltimore, MD*
Study abroad: Budapest Semesters in Mathematics, Summer 2019.

Research Interests

Commutative algebra, particularly homological and combinatorial aspects of free resolutions and related invariants; algebraic combinatorics; and noncommutative algebra. Additional interests include topological methods in algebra, computational algebra, and representation-theoretic combinatorics.

Publications

1. Noah Ripke and **Phillip Yoon**, "Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables," *Journal of Pure and Applied Algebra* **230** (2026), 108354. arXiv:2607.10639.

Research Experience

NSF REU/RET in Mathematics, California State University, Chico June–July 2026
Researcher, Commutative Algebra and Combinatorics Group *Chico, CA*

- Under the supervision of Guillermo Alesandroni, proved with a coauthor that the multigraded, graded, and total Betti numbers of monomial ideals in five variables are independent of the characteristic of the base field; led the drafting and revision of the resulting manuscript.
- Developed substantial portions of the homological and combinatorial argument using squarefree reductions, Hochster's formula, and simplicial homology; wrote Python code reducing 7,580 squarefree ideals to 210 representatives up to variable relabeling and carried out computational verification in Macaulay2.
- Continuing this work by studying characteristic-specific thresholds for Betti-number dependence through torsion in simplicial homology, including explicit eight-vertex constructions exhibiting odd-primary torsion.

Hunter College, City University of New York October 2025–present
Independent Research in Combinatorics on Words and Noncommutative Algebra *New York, NY*

- Conducting joint research with Be'eri Greenfeld on growth and nilpotent ideals in associative and monomial algebras through combinatorics on infinite words.
- Constructed a two-generated connected graded monomial algebra of finite Gelfand–Kirillov dimension with no inclusion-maximal nilpotent two-sided ideal, using combinatorial restrictions on subwords to control growth and ideal structure; continuing to investigate sharper growth constructions.

Polymath Jr., University of Minnesota June–August 2025
Research Participant *Remote*

- Investigated partition functions and recurrence operators for a metaplectic lattice model related to quantum superalgebras under the mentorship of Benjamin Brubaker.
- Contributed explicit partition-function computations and analysis of local recurrence relations and their representation-theoretic structure; presented this work at the 2026 Joint Mathematics Meetings.

Research Talks and Presentations

"Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables" 2026–2027

- **Invited talk**, AMS Special Session on Advances in Commutative Algebra and Connections with Geometry and Combinatorics, Joint Mathematics Meetings, Chicago, IL, January 2027.
- **Invited seminar talk**, CUNY Commutative Algebra and Algebraic Geometry Seminar, The Graduate Center, CUNY, New York, NY, November 2026.
- **Contributed talk**, AMS Fall Eastern Sectional Meeting, George Washington University, Washington, DC, October 2026.

- **Invited talk**, Route 81 Conference on Commutative Algebra and Algebraic Geometry, Syracuse University, Syracuse, NY, September 2026.
  Joint work with Noah Ripke.

"What Does Small Growth Mean for Algebras?" April 2026
Hunter Mathematics and Statistics Colloquium: Student Research Presentations, Hunter College, New York, NY.

"Lattice Models for Quantum Superalgebras: Color and Supercolor" January 2026
Joint Mathematics Meetings, Washington, DC.

"Finding Equations for Parametrized Curves of an Affine Variety" May 2025
Poster presentation, CUNY Directed Reading Program, The Graduate Center, CUNY, New York, NY.

Additional Mathematical Experience

CUNY Directed Reading Program, The Graduate Center January–May 2025
Algebraic Geometry and Computational Commutative Algebra *New York, NY*

- Read selected material from Cox, Little, and O'Shea's *Ideals, Varieties, and Algorithms* with graduate mentor Valeriy Sergeev; completed an expository project and final presentation.

Teaching Experience

Hunter College, City University of New York August 2026–present
Adjunct Lecturer *New York, NY*
Instructor of record: Precalculus (MATH 12550).

Columbia Grammar and Preparatory School August 2025–August 2026
Teacher of Mathematics *New York, NY*
Courses: AP Calculus AB, Precalculus.

New York City College of Technology, CUNY August–December 2024
Adjunct Lecturer *Brooklyn, NY*
Courses: College Algebra and Trigonometry, Precalculus.

Trinity School August 2022–June 2024
Mathematics Teaching Fellow *New York, NY*
Courses: Prealgebra, Honors Algebra I.

Johns Hopkins University August 2018–May 2021
Teaching Assistant and Tutor *Baltimore, MD*
Courses: Discrete Mathematics, Probability and Statistics, Calculus II, Linear Algebra.

Awards and Honors

- John Arthur Loustau Scholarship, Hunter College, CUNY, 2026.
- Dr. Aurel Wintner Scholarship and Gail J. McGovern Scholarship, Johns Hopkins University, 2018–2020.
- Honorable Mention, International Olympiad on Astronomy and Astrophysics, 2016.

Service and Outreach

- Member, AI Committee, Columbia Grammar and Preparatory School, 2025–2026.
- MATHCOUNTS Coach, Trinity School, 2022–2024.
- Undergraduate Representative, Krieger School of Arts and Sciences Curriculum Committee, Johns Hopkins University, 2020–2021.
- Co-organizer, Johns Hopkins Mathematics Tournament, 2018–2020.
- Exam Writer and Supervisor, U.S. Astronomy and Astrophysics Olympiad (2017) and Maryland Science Olympiad (2018–2019).

Languages and Technical Skills

Languages: English, Korean, Japanese (JLPT N1).
Technical: LaTeX, Python, Macaulay2.

Additional Professional Experience

xAI January–August 2025
AI Tutor *Remote*

- Evaluated and improved AI-generated mathematical reasoning and written solutions.